# Daily treated-effluent volume forecasting with limited local history: development baselines and feasibility

Development work draft: authors and affiliations pending

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**Scope notice.** This is a reproducible development report, not a complete final-test journal manuscript. Shared models and prediction intervals have not been evaluated. Site identifiers are not yet verified independent physical facilities, and historical first-arrival/revision times are unavailable. The final 2024–2025 forecasting evaluation is pending.

**Abstract**

We examine the feasibility of forecasting daily treated-effluent discharge volume under limited initial local history using an explicitly exploratory cohort from a released Wessex Water data snapshot. Ten site identifiers were selected using 2016–2021 coverage before model outputs were inspected. Persistence, weekly persistence and fixed local ridge regression were evaluated in four 2022–2023 episodes with 30-, 90- and 180-day calendar-history budgets and assumed extra observation delays of zero, one and two days. For zero extra delay and a 90-day budget, the equal-site mean MAE skill of local ridge against persistence was -31.88% and -33.88% for one- and three-day horizons. These negative findings do not establish transfer-learning effectiveness. Physical-facility mapping, genuinely available historical observations, shared-model comparisons and final evaluation remain necessary before engineering or cross-facility claims are justified.

## 1 Research question and evidence boundary

The intended research question is whether shared learning can improve one- and three-day forecasts of daily treated-effluent discharge volume at existing wastewater facilities excluded from shared training, while reducing the required initial local modeling history. Limited-history truncation represents an existing facility with insufficient available modeling records, not startup of a newly constructed treatment plant. These volumes alone cannot demonstrate effluent-quality compliance, pollutant loading, overflows, process optimization or energy savings.

Prior publications explicitly address effluent prediction under insufficient data and transfer learning (Yu et al., 2024; Huang et al., 2023); their full methods have not yet been independently reviewed for this project. A publisher full-text section review confirms prior bidirectional CNN/LSTM transfer for discharge and BOD at two industrial wastewater plants (Kim et al., 2024). That study uses daily outlet monitoring observations from 2018–2022, removes entire records with missing parameters or specified outliers, and reports a 7:3 training/test ratio. Its discharge physical units, forecast lead, split ordering and inclusive target adaptation budget remain unspecified in the reviewed methods. Neither a first-transfer claim nor an assertion that it duplicates this exact daily-volume protocol is warranted.

Probabilistic wastewater inflow forecasting using rainfall forecasts and sewer sensors has also been studied (Sonnenschein and Ziel, 2023), and sewer-flow/water-level GPR forecasting provides additional uncertainty-related context (Rezaee et al., 2025). Inflow, water level and treated-effluent

daily volume are distinct targets. Here, simple local baselines are evaluated before adopting shared or complex models. The secondary planned questions concern negative transfer under limited history and missing/delayed inputs, and empirical interval coverage versus width. Those experiments are pending.

# 2 Executed development methods

## 2.1 Data version, target and audit

The supplied Wessex Water dataset covers 2016–2025 (Wessex Water, 2026). The raw CSV SHA256 is:

```
22f494e2b96fe6fa9550c68971d50c752ddd291793fdc60334d64b2ef089f518
```

All 968794 raw records were independently checked against 31 saved API pages. Names containing WATER RECYCLING CENTRE define 936637 records and 261 site identifiers. Provider metadata defines TDV at WRC/WTC permitted outfalls as effluent discharged to the environment; the recorded unit is $m^3$ per calendar-day total. Pumping stations and water-treatment works are excluded.

The primary retrospective target rule accepts Good, nonnegative, nonconflicting values. Invalid targets remain absent on the full daily calendar; no target interpolation is performed. The supplied positive-only/no-duplicate rule yields 920445 records; retaining 93 Good zeros yields 920538 primary candidates. All originals and conflicting records are preserved. WRC records include 534 Good negative values, 396 Missing nonzero values and two conflicting station-day keys. These counts are descriptive audit results, not model findings. Meter certification is not universal, so a universal instrumental uncertainty or prediction-interval guarantee is inappropriate.

## 2.2 Cohort, periods and offline timing scenarios

The cohort rule, stored before the first real model output, requires at least 95% calendar-day coverage by Good, nonnegative, nonduplicated records in each year 2016–2021 and a single provided name per identifier. It yields 199 candidate identifiers. Numerical sorting reserves the first three as possible future sources; the next ten are the development identifiers. The demonstration list of 50 sites is not reused. No source model was trained in this stage and physical independence is unverified.

Four first day-end forecast origins were fixed: 1 January and 1 July in 2022 and 2023, each followed by 60 calendar-day origins. An origin $o$ predicts $Y_{o+h}$ for $h \in \{1, 3\}$; equivalently, the issue date $t = o + 1$ predicts $Y_t$ and $Y_{t+2}$, not a three-day sum. Input observations are allowed through $o - d$, with assumed extra delay $d \in \{0, 1, 2\}$.

Public snapshots do not authenticate first operational availability or historical revisions. Consequently these are retrospective scenarios, not live deployment tests. Finite nonnegative raw values form the assumed input and fitting-label stream without retrospective QA masking; conflict days are quarantined. This may retain Missing-coded zeros in that assumed stream. Final QA is used only for retrospective evaluation truth. Fixed delay does not repair revision bias.

## 2.3 Inclusive history budgets and baselines

For first origin $T$, initial history is the calendar window $[T - B, T - 1]$, with $B \in \{30, 90, 180\}$. Features, scaling, fitting and any future adaptation/calibration must lie within this history; gaps

count toward the budget. Training labels additionally satisfy target date $\leq T - 1 - d$. During an episode parameters remain fixed while newly arrived assumed observations may update lags. There is no history warm-up before $T - B$.

Persistence takes the most recent allowed finite observation, searching backward at most seven additional days when that day's value is missing. Weekly persistence uses the exact target date minus seven days, if already available; it does not synthesize missing values. Local ridge uses volume lags $0, 1, 2, 3, 6, 7$ relative to $o-d$, the past seven-day observed mean and sine/cosine encoding of the target weekday. Median imputation, available missing indicators and feature standardization are fit only on the training subset. Ridge regularization was fixed at $\alpha = 10$ without tuning. At least 14 training labels are required. Negative model outputs are retained, not clipped after observing performance.

### 2.4 Evaluation and verification

All three models use identical valid target/origin dates within each condition; equality across budget curves was independently checked. MAE and RMSE are computed per identifier after pooling the four prescribed episodes, then summarized with equal identifier weights. Relative skill is $1 - \mathrm{MAE}_{\mathrm{model}}/\mathrm{MAE}_{\mathrm{baseline}}$ per identifier, followed by equal averaging; zero-baseline skill is undefined. Ratios of aggregate MAEs are not substituted for mean per-site skill.

The 2024–2025 records were read only for disclosed descriptive audit, not forecast performance, tuning or selection. Development code rejects evaluation labels in those years and truncates modeling input records to 2023. Software fixtures and real-data validation are separate; checks cover dates, arrival bounds, inclusive budgets, training-label boundaries, common dates, hashes and truth matching. No day-wise independence test or facility-population significance claim is made.

## 3 Verified development findings

All 12 planned full development runs completed; the saved prediction grid contains 129600 model-specific rows. An independently run two-site interface episode reproduced the corresponding full-run predictions exactly and was excluded from the main aggregate. Each identifier/condition has 227–240 common valid targets across at most 240 scheduled origins. The three models predict all available valid targets; this conditional coverage does not imply 100% calendar-day truth coverage.

Table 1: Zero-extra-delay development MAE ($m^3$) and equal-site relative skill. This is an exploratory SiteID cohort, not a final physical-facility test.

| Budget | Horizon | Persistence | Weekly | Local ridge | Ridge skill |
|---|---|---|---|---|---|
| 30 | 1 | 2012.04 | 3790.81 | 4717.78 | -99.50% |
| 30 | 3 | 3101.60 | 3782.58 | 6797.93 | -108.76% |
| 90 | 1 | 2012.04 | 3790.81 | 2755.57 | -31.88% |
| 90 | 3 | 3101.60 | 3782.58 | 4155.91 | -33.88% |
| 180 | 1 | 2012.04 | 3790.81 | 2352.98 | -19.92% |
| 180 | 3 | 3101.60 | 3782.58 | 3546.50 | -17.59% |

At zero extra delay, 90-day local ridge is worse than persistence at all ten identifiers for both horizons. Even with 180-day history, mean skill remains negative, with nine and eight identifiers worse for one- and three-day horizons. Average local-ridge skill is negative in every tested budget/horizon/delay condition. Longer initial history reduces the observed average error but does not

establish parity with persistence.

There are 687 negative-volume forecasts among 43200 local-ridge prediction rows; all occur under the 30-day budget. These outcomes remain in the archived predictions and errors. They are model failures, not modified observations or measured negative-transfer prevalence: no shared model has yet been evaluated. Figure 1 displays all site ratios; Figure 2 retains the poor short-budget delay case.

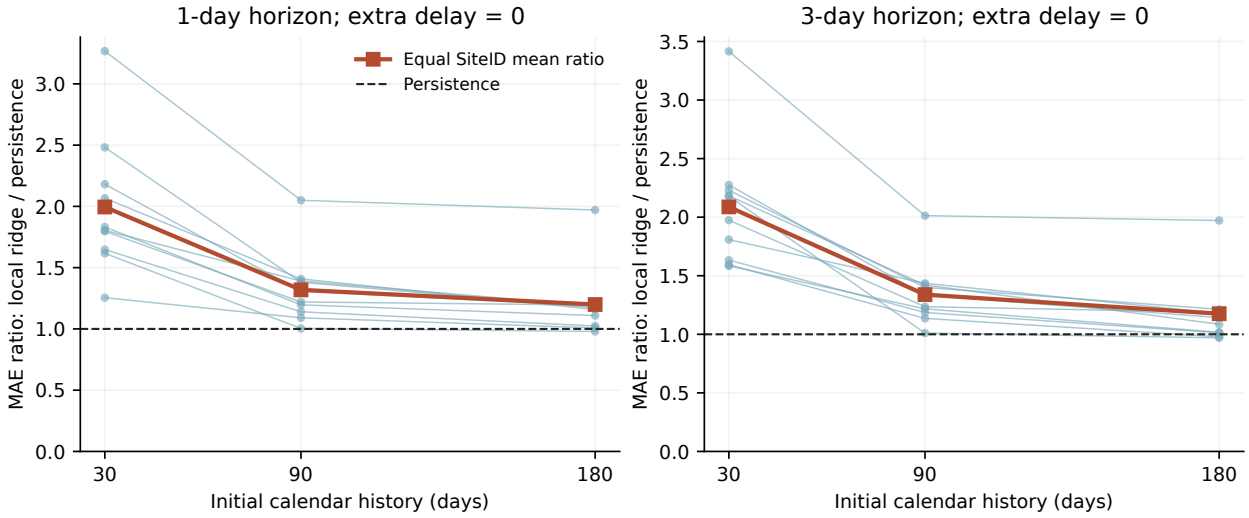



Figure 1: Local-ridge MAE relative to persistence across inclusive initial calendar-history budgets, at zero assumed extra delay. Thin curves are the ten exploratory site identifiers; the thick curve is their equal-weight mean ratio. Values above one indicate worse error. Results pool only the four predefined development episodes.

## 4 Feasibility decision, limitations and pending experiments

The present fixed local-ridge specification does not improve on persistence. Short-window estimation instability, distribution drift and fixed-parameter adaptation limits are plausible explanations, not established causal findings. No model, site or cleaning rule was replaced to obtain positive results. Persistence should remain a strong comparator in the next development stage; beating only weekly persistence would not establish utility.

A limited shared-ridge and budget-contained bias-correction pilot is a reasonable next test of whether external estimation can stabilize short histories. This is a decision to investigate, not evidence that sharing works. Source/development identifier disjointness is insufficient for a strict physical-facility transfer claim. Facility/permit/outfall mapping and historical availability evidence remain unresolved. Calibration and adaptation must be separated chronologically, and interval assessment remains pending.

**Incomplete manuscript items:** authenticated full-text comparison; current official journal scope/author requirements; final physical-facility cohort and frozen protocol; shared-model and negative-transfer experiments; dependence-aware uncertainty estimates; missing-input and interval experiments; final 2024–2025 evaluation; true authorship, affiliations, funding and disclosure statements. No final conclusions or journal submission claims are made. Hong Kong flow definitions remain incomparable and Thames raw files are unavailable.

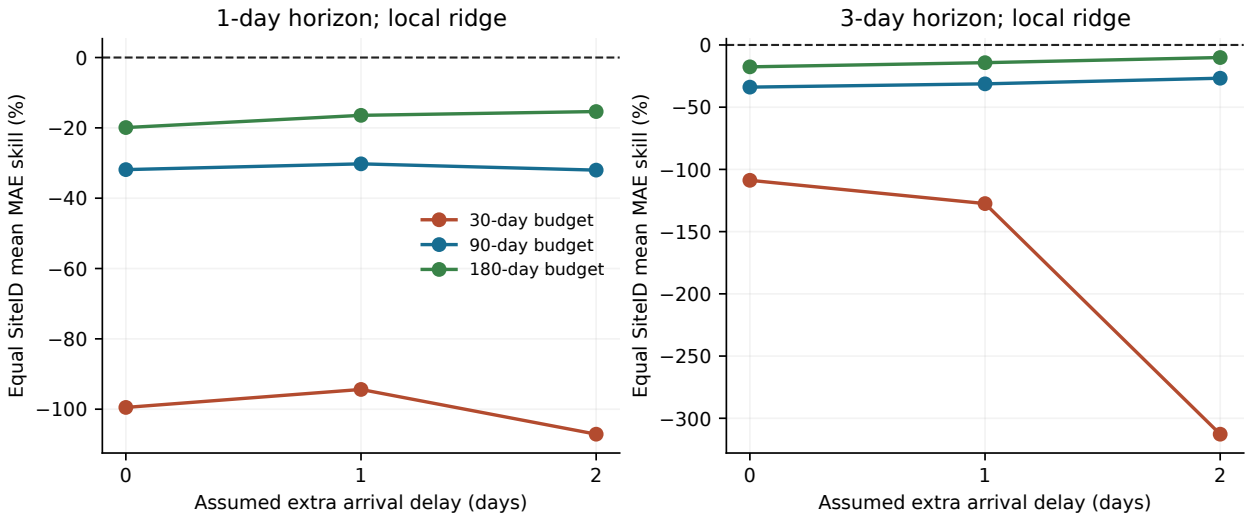



Figure 2: Equal-site mean MAE skill of fixed local ridge versus persistence under assumed extra observation delays. Negative values indicate worse error. These are historical-snapshot timing scenarios, not authenticated operational delays or independent physical-facility inference.

**Data and code.** Wessex data are attributed to Wessex Water under the supplied CC BY 4.0 metadata. Raw input bytes, cleaning/audit provenance, configurations, predictions and result/figure hashes are recorded in the companion development package. Original data are intentionally excluded from the Overleaf ZIP. This draft's numbers are generated from checked saved result files; it has not been synchronized to an account or publicly submitted.

## References

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